HUMAN API

A governed workflow for work questions when an expert is unavailable

Eureka Innovation Hackathon · Future of Work @ Presight

Early concept · local demo · fictional data

A clear, safe way to handle work questions when the expert is unavailable.

What is Human API?

A colleague chooses what questions the software may help with and for how long. The person asking must still be allowed to open every file used. The app shows what a source says and what it does not say, or routes the question to a human.

Example: Alice asks why a timeout changed from 90 to 120 seconds. Bob opted in for this project. The app finds a written decision Alice can read. It explains why the change was approved for release 5.2, but cannot tell whether the setting is still needed today. The answer includes the source and that limitation; it is not presented as Bob's reply.

This is an early local demo with fictional people and files. It finds evidence by matching words and uses fixed response templates; it does not use a language model. Microsoft sign-in and file-search adapters are untested scaffolding, and no Teams bot exists. See the accompanying Reader Guide for a glossary and AI-agent instructions.

Submission concept and synthetic-data MVP · 26 October 2026 · Dubai

Problem, people and scenario

The workplace dependency problem

Work stalls when an employee needs a decision rationale, operational detail or project history held by a colleague who is busy, offline, on leave or no longer assigned. The knowledge may already be in an ADR, runbook or approved project source, but finding and validating it still takes active time. Waiting time is not automatically productive time saved.

Fictional scenario

Alice, a platform engineer, asks Bob why Project Alpha's startup timeout changed to 120 seconds and whether the change is still needed. Bob is on fictional leave and has approved a time-limited policy for this project, this topic, selected files and named coworkers. ADR-024 is a written technical decision record. It explains that the change was approved for release 5.2 because dependency setup sometimes took more than 90 seconds. It does not say whether 120 seconds is still needed. Human API shows this partial answer with the source and limitation; it does not speak as Bob.

Users

- Requester: submit a question, view an answer and evidence, give feedback or request review.
- Knowledge owner: opt in, choose dates, projects, sources, topics, audience and backup; revoke at any time.
- Manager: approve scope changes and monitor team blocking patterns without unrestricted access to private question content.
- Security administrator: manage policy and audit metadata; no default ability to read every question or source.

Solution and workflow

Lifecycle

- Save the question, who asked it, who may know, the project and the response deadline.
- Wait for the agreed response time, unless an owner-approved out-of-office rule applies. Busy status alone never triggers the helper.
- Check the owner's limits: dates, project, topic, approved coworkers, files and manager review.
- Before searching a file, check that the person asking can already open it.
- Give a source-backed answer and its limits, deny a restricted request without exposing details, or send a question to a human.
- Label a human reply as human. Require approval before reusing a correction.

MVP screens

The React app contains a dashboard, ask flow, My AI Delegate controls, escalation inbox and governance view. It uses real local API results. Identity and availability are explicitly synthetic. The interface labels software-produced text as AI-generated to distinguish it from a later human response; the demo uses fixed templates, not a language model.

Safety outcome: lack of evidence is a reason to abstain. It is not an invitation to fill gaps with plausible text.

Architecture and technical feasibility

Developer view: the MVP is a modular monolith—one app organized into separate code modules. The web API calls rule-based policy, access, search and workflow modules. SQLAlchemy saves people, projects, policies, files, questions, status history, deadlines, answers, citations, escalations, reviews and audit events in a relational database.

A database-backed timeout scanner polls pending jobs; conditional transitions and unique question/job/answer relationships reduce duplicate outcomes in the single-process demo. Production multi-worker operation still needs database-specific claims/locks, an outbox, observability, recovery tests and idempotent delivery.

The demo finds files by matching words; it does not understand paraphrases or meaning as a language model would. It checks the requester's project membership and file-access list before scoring results. It cites matching text, identifies known gaps or conflicts, and stops when evidence is not enough. It does not use embeddings, a vector database or a language model. This is a deliberate demo limit, not a recommended production search engine.

Technical stack

Python 3.12, FastAPI 0.162.0, Pydantic 2.14.1, SQLAlchemy 2.1.6, Alembic 1.20.1, PostgreSQL 18.6 for Compose, psycopg 3.3.6, React 19.3.1, TypeScript 7.0.2 and Vite 8.3.5. Python versions are pinned; npm lockfiles are included. No model weights or paid AI APIs are bundled.

LangGraph was not selected: a small deterministic state machine with human handoff and durable rows is sufficient at this scope. Add orchestration only if a measured multi-tool/checkpoint need appears.

Microsoft 365 integration

Recommended path: build a Teams or Copilot Studio extension around this question workflow. Reuse Microsoft work sign-in and file permissions instead of building a new company-wide search product.

- Work sign-in: early Microsoft Entra ID token-checking code exists. It needs an app registration and tenant testing before it can sign anyone in.
- SharePoint: an early Microsoft Graph adapter is designed to search and fetch files as the person asking, preserving that person's file permissions. It has not been tested end to end in a tenant.
- Outlook: a possible adapter reads only the signed-in owner's out-of-office status and dates, then discards the reply text. It does not read inbox messages.
- Teams: no bot is implemented. A future Teams app needs registration, an internet-reachable secure endpoint and permission to install the app in the tenant.

Microsoft Graph uses named permissions to control which files an app may search. Some permission choices need administrator approval. Choose the narrowest permission that works in the target tenant. Never use the owner's broader file access to answer someone else's question. Permission, licensing and tenant rules must be tested before a pilot.

Security, privacy and responsible AI

Non-negotiable policy

- If identity or permission is unclear, deny the request. Require verified work sign-in and the owner's explicit approval or another approved organizational policy.
- The owner's approval never gives another person new file access. Keep the requester's file permissions and check again before showing saved evidence.
- Restricted files need a named list of people who may open them; being on the project alone is not enough.
- Do not reveal hidden titles/snippets through denials, summaries, caches or admin views. Keep security administration separate from content access.
- Treat file contents as information, not commands. The demo filters a few known hostile phrases; a real service needs broader attack testing and carefully limited tools.
- Never impersonate the owner. Mark AI text as generated. Human corrections require publication approval.

Synthetic labels are examples; they are not Presight's official sensitivity categories. The app does not include encrypted database keys, production data deletion rules, data-loss-prevention labels, incident response or security certification. Use fictional data until a security/privacy review and pilot controls are complete.

Competitive position

Microsoft 365 Copilot, Copilot Studio, SharePoint agents, Glean and Moveworks already offer enterprise search or grounded answers, citations, permission-aware sources, configurable agents and/or human handoff. Microsoft announced Autopilot (formerly Scout) on 25 September 2026 as a persistent proactive agent with a tenant identity; the announcement said private preview was expanding at the end of September, not general availability. Microsoft also announced a Copilot/agent-curated team knowledge base in SharePoint Frontier Preview on 14 October, while Copilot in SharePoint began its GA rollout on 30 September. This substantially overlaps broad “knowledge continuity” messaging.

A plausible differentiating workflow is narrower: an employee-consented and dated delegation; unanswered request with a configured deadline; project/topic/audience/source scope; requester-side live ACL; evidence-backed answer or abstention; backup escalation; approved correction history. Public materials do not prove that this exact lifecycle is unique. Validate it against Microsoft configurability and customer interviews.

Recommendation

Prototype inside Teams or Copilot Studio first. Use SharePoint agents or Microsoft 365 Copilot for general knowledge grounding; use Human API only for explicit handoff, response-window, approval and audit needs. Reassess against Autopilot preview and vendor capabilities before any standalone investment.

Glean and Moveworks do not publish comparable public list prices in the reviewed technical sources. Microsoft Copilot/Studio/agent usage depends on eligible user plans, capacity or usage-based billing and tenant setup. Procurement requires current vendor quotes and tenant-specific licensing review.

Implemented MVP and evaluation

What works today

- 20 fictional employees, 4 teams, 5 projects, 55 approved synthetic documents and a demo policy set.
- Persistent questions, timeout jobs, owner consent, manager approval, scope checks, evidence citations, denial, abstention, correction and audit metadata.
- 17 automated tests pass, including restart durability and idempotency checks; see Evaluation Report for coverage.
- 36 labeled requester-scoped retrieval cases: 32 expected authorized hits and 4 expected no-evidence cases. All 36 pass; this is lexical retrieval/ACL fixture coverage only, not answer accuracy or security assurance.

Covered tests include prompt-injection fixture handling, denial without restricted title/snippet, policy revocation, forged demo identity in enterprise mode, owner approval, deadline cancellation, retry limits and human-before-timeout race. No live Microsoft API, LLM evaluation, performance benchmark or real-user pilot has been run.

MVP acceptance boundary

The local app runs with documented commands and synthetic data. Entra/Graph calls need tenant credentials and consent; the enterprise question route remains disabled. Teams messaging, production token management, persistent cache invalidation, multi-instance job leasing and calibrated semantic validation remain future work.

Business value and illustrative time estimate

The hoped-for benefit is less active time spent searching, asking again and reconstructing decisions. Count only work time a pilot can show was actually avoided after a useful answer. Do not count hours spent waiting as saved work by default.

Example math, not a Presight measurement: 500 people × 2 blocking questions each week × 50 weeks = 50,000 possible questions a year. The scenarios below then estimate what share is suitable, used and resolved, and how many active minutes are saved. The AED 250 hourly labor rate is also an assumption.

Scenario	Suitable × used × solved	Active min saved	Time value / year
Low	30% × 40% × 50% = 3,000	10	AED 125,000
Base	40% × 60% × 70% = 8,400	20	AED 700,000
High	55% × 75% × 75% ≈ 15,469	25	AED 1,611,328

Illustrative annual operating cost: AED 740,000 before added Microsoft licenses. This assumes 0.7 of a full-time engineer (AED 336k), 0.4 of a full-time operations/support role (AED 192k), infrastructure and monitoring (AED 60k), security/privacy review (AED 120k), and adoption/support (AED 32k). These are planning placeholders, not quotes. The base estimate of time value is AED 40k below the estimated operating cost, so a pilot must prove value or the cost must fall.

Potential future pricing could be per user/team, private deployment, custom connectors and support. Price and revenue are hypothetical; no willingness-to-pay evidence exists. M365/agent licensing cost is excluded because it depends on the tenant contract.

Risks, roadmap and references

Risks and mitigations

- Showing a file to someone who lost access: check requester permissions before search and again before displaying saved evidence; test revoked access.
- Incorrect or stale answer: cite source version/date, show uncertainty, abstain on conflict, require expert approval for reuse.
- Employee surveillance concern: explicit opt-in, no inbox reading, no Busy-as-consent, date-bound revocation.
- Weak business case or competing Microsoft features: run a short team pilot, compare with existing Copilot setup, and stop if users do not save active time or trust the workflow.

Next steps

0–2 weeks: security/privacy and product discovery; choose one low-sensitivity team. 2–6 weeks: Entra + delegated Graph proof and Teams workflow, with source permissions revalidated. 6–10 weeks: pilot against labeled questions and human acceptance, tracking active work minutes, coverage, correction, escalation and zero access violations. Advance only with approval from Security, Privacy, Microsoft platform owners and pilot users.

Selected references (accessed 26 October 2026)

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